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高数B上. HW5.

习题 3.1.

4. $\int (2x^{3/2}+1)^{2/3}\sqrt{x}\,dx$ 由于 $d(2x^{3/2}+1) = 2\sqrt{x}\cdot\frac{3}{2}dx = 3\sqrt{x}\,dx$.

令 $y = 2x^{3/2}+1$. 原式 $= \frac{1}{3}\int y^{2/3}\,dy = \frac{1}{5}y^{\frac{5}{3}}+C = \frac{1}{5}(2x^{3/2}+1)^{\frac{5}{3}}+C$.

5. 令 $y = \frac{1}{x}$. $dy = -\frac{dx}{x^2}$

$$\int \frac{e^{\frac{1}{x}}}{x^2}dx = -\int e^y dy = -e^y + C = -e^{\frac{1}{x}} + C$$

7. $$\int \frac{dx}{3+5x^2} = \frac{1}{5}\cdot\int \frac{dx}{x^2+\frac{3}{5}} = \frac{1}{5}\cdot\sqrt{\frac{5}{3}}\arctan\sqrt{\frac{5}{3}}x + C = \frac{1}{\sqrt{15}}\arctan\sqrt{\frac{5}{3}}x + C$$

9. $d\sqrt{x} = \frac{dx}{2\sqrt{x}}$

故 $$\int \frac{dx}{\sqrt{x}(1+x)} = 2\int \frac{d\sqrt{x}}{1+(\sqrt{x})^2} = 2\arctan\sqrt{x} + C.$$

12. $$\int \frac{dx}{e^x - e^{-x}} = \int \frac{e^x dx}{e^{2x}-1} = \int \frac{de^x}{(e^x)^2-1} = \frac{1}{2}\ln\left|\frac{e^x-1}{e^x+1}\right| + C.$$

15. $\int \frac{dx}{1-\sin x}$ $\left(1-\sin x = \cos^2\frac{x}{2}+\sin^2\frac{x}{2}-2\sin\frac{x}{2}\cos\frac{x}{2} = (\cos\frac{x}{2}-\sin\frac{x}{2})^2\right)$

$$= \int \frac{dx}{(\cos\frac{x}{2}-\sin\frac{x}{2})^2} = \int \frac{dx}{2\sin^2(\frac{x}{2}-\frac{\pi}{4})} = \int \frac{d(\frac{x}{2})}{\sin^2(\frac{x}{2}-\frac{\pi}{4})} = -\cot(\frac{x}{2}-\frac{\pi}{4})+C$$

13. $$\int \frac{\ln\ln x}{x\ln x}dx = \int \frac{\ln\ln x}{\ln x}d\ln x = \int \ln\ln x\, d\ln\ln x = \frac{1}{2}(\ln\ln x)^2 + C.$$

14. $$\int \frac{dx}{1+\cos x} = \int \frac{dx}{1+2\cos^2\frac{x}{2}-1} = \int \frac{d(\frac{x}{2})}{\cos^2\frac{x}{2}} = \tan\frac{x}{2} + C.$$

17. $$\int \frac{x^{2n-1}}{x^n-1}dx = \frac{1}{n}\int \frac{x^n}{x^n-1}d(x^n) = \frac{1}{n}\int (1+\frac{1}{x^n-1})\,dx^n$$

$$= \frac{x^n}{n} + \frac{1}{n}\ln|x^n-1| + C.$$

20. $$\int \frac{e^{\arctan x}+x\ln(1+x^2)}{1+x^2}dx = \int \frac{e^{\arctan x}}{1+x^2}dx + \int \frac{x\ln(1+x^2)}{1+x^2}dx$$

$$= \int e^{\arctan x}\,d\arctan x + \frac{1}{2}\int \frac{\ln(1+x^2)}{1+x^2}\,d(1+x^2)$$

$$= e^{\arctan x} + C + \frac{1}{2}\int \ln(1+x^2)\,d\ln(1+x^2) = e^{\arctan x} + \frac{1}{4}(\ln(1+x^2))^2 + C.$$

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22. $\int \sin^2\frac{x}{2}\cos\frac{x}{2}\,dx$ $\left(d\sin\frac{x}{2} = \frac{1}{2}\cos\frac{x}{2}\,dx\right)$.

$= 2\int \sin^2\frac{x}{2}\,d\sin\frac{x}{2} = \frac{2}{3}\sin^3\frac{x}{2} + C$.

26. $\int \frac{dx}{(a^2-x^2)^{3/2}}$ 令 $x = a\cos\theta$. $dx = -a\sin\theta\,d\theta$. $\sqrt{a^2-x^2} = a\sin\theta$.

$\theta \in [0,\pi]$.

$\cot\theta = \frac{\cos\theta}{\sin\theta} = \frac{x}{\sqrt{a^2-x^2}}$

$= \int \frac{-a\sin\theta\,d\theta}{a^3\sin^3\theta} = -\frac{1}{a^2}\int \frac{d\theta}{\sin^2\theta} = \frac{1}{a^2}\cot\theta + C = \frac{1}{a^2}\frac{x}{\sqrt{a^2-x^2}} + C$

27. $\int \frac{\sqrt{x^2-a^2}}{x}\,dx$. $(a>0)$. 令 $x = a\sec\theta$. $\Rightarrow \theta = \arccos\left(\frac{a}{x}\right)$

(i) $x \geq a$ 时, $\theta \in [0, \frac{\pi}{2})$. $\sqrt{x^2-a^2} = a\tan\theta$. $dx = a\sec\theta\tan\theta\,d\theta$.

原式 $= \int \frac{a\tan\theta}{a\sec\theta}\cdot a\sec\theta\tan\theta\,d\theta = a\int \tan^2\theta\,d\theta$

$= a\int \frac{1-\cos^2\theta}{\cos^2\theta}\,d\theta = a\int(\sec^2\theta - 1)\,d\theta = a\tan\theta - a\theta + C$.

$= \sqrt{x^2-a^2} - a\arccos\frac{a}{x} + C$.

(ii) $x \leq -a$ 时, $\theta \in (\frac{\pi}{2}, \pi]$. $\sqrt{x^2-a^2} = -a\tan\theta$.

$\Rightarrow$ 原式 $= -a\tan\theta + a\theta + C = \sqrt{x^2-a^2} + a\arccos\frac{a}{x} + C$.

29. $\int \frac{dx}{\sqrt{1+e^{3x}}}$ 令 $t = \sqrt{1+e^{3x}}$. $x = \frac{1}{3}\ln(t^2-1)$ $\Rightarrow dx = \frac{2}{3}\frac{t}{t^2-1}\,dt$

$= \int \frac{2t\,dt}{3t(t^2-1)} = \frac{2}{3}\int \frac{dt}{t^2-1} = \frac{1}{3}\ln\left|\frac{t-1}{t+1}\right| + C$

$= \frac{1}{3}\ln\frac{\sqrt{1+e^{3x}}-1}{\sqrt{1+e^{3x}}+1} + C = \frac{1}{3}\ln\frac{(\sqrt{1+e^{3x}}-1)^2}{1+e^{3x}-1} + C = \frac{2}{3}\ln(\sqrt{1+e^{3x}}-1) - x + C$.

31. 令 $x = \tan\theta$. $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$. $\sqrt{x^2+1} = \frac{1}{\cos\theta}$ $dx = \frac{d\theta}{\cos^2\theta}$

$\Rightarrow \int \frac{dx}{x^6\sqrt{1+x^2}} = \int \frac{\sec^2\theta\,d\theta}{\tan^6\theta\sec\theta} = \int \frac{\cos^5\theta\,d\theta}{\sin^6\theta} = \int \frac{(1-\sin^2\theta)^2}{\sin^6\theta}\,d\sin\theta$

$= \int \frac{1}{\sin^6\theta} - \frac{2}{\sin^4\theta} + \frac{1}{\sin^2\theta}\,d\theta = -\frac{1}{5\sin^5\theta} + \frac{2}{3\sin^3\theta} - \frac{1}{\sin\theta} + C$.

$\left(\sin\theta = \frac{x}{\sqrt{1+x^2}}\right)$. $= -\frac{1}{5}\frac{(1+x^2)^{5/2}}{x^5} + \frac{2}{3}\frac{(1+x^2)^{3/2}}{x^3} - \frac{\sqrt{1+x^2}}{x} + C$.

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34. $\int \sqrt{7+x-x^2}\,dx$ $\quad 7+x-x^2 = \frac{29}{4} - (x-\frac{1}{2})^2$

令 $y = x-\frac{1}{2}$. 上式 $= \int \sqrt{\frac{29}{4}-y^2}\,dy$.

用解决一般的 $\int \sqrt{a^2-y^2}\,dy$. 令 $y = a\cos\theta$. $\theta \in [0,\pi]$

$\sqrt{a^2-y^2} = a\sin\theta$. $dy = -a\sin\theta\,d\theta$

$= -a^2\int \sin^2\theta\,d\theta = \frac{a^2}{2}\int(\cos 2\theta - 1)\,d\theta$

$= \frac{a^2}{4}\sin 2\theta - \frac{a^2}{2}\theta + C$.

$= \frac{a^2}{2}\sin\theta\cos\theta - \frac{a^2}{2}\theta + C = \frac{1}{2}y\sqrt{a^2-y} - \frac{a^2}{2}\arccos\frac{y}{a} + C$.

代入 $a = \sqrt{\frac{29}{4}}$.

原式 $= \frac{1}{2}y\sqrt{\frac{29}{4}-y^2} - \frac{29}{8}\arccos\frac{2y}{\sqrt{29}} + C$. 代回 $y = x-\frac{1}{2}$

$= \frac{2x-1}{4}\sqrt{7+x-x^2} - \frac{29}{8}\arccos\frac{2x-1}{\sqrt{29}} + C$.

习题3.2.

5. $\int \arctan x\,dx = x\arctan x - \int x\,d\arctan x$

$= \int \frac{x}{1+x^2}\,dx = \frac{1}{2}\int \frac{d(1+x^2)}{1+x^2} = \frac{1}{2}\ln(1+x^2) + C$

$= x\arctan x - \frac{1}{2}\ln(1+x^2) + C$.

8. $\int e^{ax}\sin bx\,dx$ 配以对偶 $\int e^{ax}\cos bx\,dx$.

猜原函数是 $e^{ax}(A\sin bx + B\cos bx)$.

求导: $ae^{ax}(A\sin bx + B\cos bx) + e^{ax}(Ab\cos bx - Bb\sin bx)$

$= e^{ax}\cdot((aA-Bb)\sin bx + (aB+Ab)\cos bx) = e^{ax}\sin bx$

故 $\begin{cases} aA - bB = 1 \\ bA + aB = 0. \end{cases} \Rightarrow \begin{cases} A = \frac{a}{a^2+b^2} \\ B = \frac{-b}{a^2+b^2}. \end{cases}$

故 $\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx) + C$.

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11. $\int \ln(x+\sqrt{1+x^2})\,dx = x\ln(x+\sqrt{1+x^2}) - \int x\, d\ln(x+\sqrt{1+x^2})$

$\int x\, d\ln(x+\sqrt{1+x^2}) = \int \frac{x}{\sqrt{1+x^2}}\,dx = \frac{1}{2}\int \frac{d(1+x^2)}{\sqrt{1+x^2}} = \sqrt{1+x^2}+C$

原式 $= x\ln(x+\sqrt{1+x^2}) - \sqrt{1+x^2}+C.$

12. $\int \arccos^2 x\,dx = x\arccos^2 x - \int x\, d\arccos^2 x$

$\int x\, d\arccos^2 x = 2\int x\arccos x \frac{-1}{\sqrt{1-x^2}}\,dx = 2\int \arccos x\, d\sqrt{1-x^2}.$

$= 2\sqrt{1-x^2}\arccos x - 2\int \sqrt{1-x^2}\, d\arccos x$

$= 2\sqrt{1-x^2}\arccos x - 2\int \frac{\sqrt{1-x^2}\cdot(-1)}{\sqrt{1-x^2}}\,dx$

$= 2\sqrt{1-x^2}\arccos x + 2x + C$

$\Rightarrow$ 原式 $= x\cdot\arccos^2 x - 2\sqrt{1-x^2}\arccos x - 2x + C.$

15. $\int \frac{\arcsin x}{x^2}\,dx = -\int \arcsin x\, d\frac{1}{x} = -\frac{\arcsin x}{x} + \int \frac{1}{x}\, d\arcsin x$

$= -\frac{\arcsin x}{x} + \int \frac{dx}{x\sqrt{1-x^2}}$

$\int \frac{dx}{x\sqrt{1-x^2}} \overset{x=\cos\theta}{=\!=} \int \frac{-\sin\theta\, d\theta}{\cos\theta\sin\theta} = -\int \frac{d\theta}{\cos\theta} = -\int \frac{\cos\theta\, d\theta}{\cos^2\theta}$

$= \int \frac{d\sin\theta}{1-\sin^2\theta} = \frac{1}{2}\ln\frac{1-\sin\theta}{1+\sin\theta} + C = \frac{1}{2}\ln\frac{1-\sqrt{1-x^2}}{1+\sqrt{1-x^2}} + C$

$= \ln(1-\sqrt{1-x^2}) - \ln|x| + C$

$\Rightarrow$ 原式 $= -\frac{\arcsin x}{x} + \ln(1-\sqrt{1-x^2}) - \ln|x| + C.$

16. $\int x^3(\ln x)^2\,dx = \frac{1}{4}\int \ln^2 x\, dx^4 = \frac{1}{4}x^4\ln^2 x - \frac{1}{4}\int x^4\, d\ln^2 x$

$\int x^4 d\ln^2 x = 2\int x^4\ln x\cdot\frac{1}{x}\,dx = 2\int x^3\ln x\,dx = \frac{1}{2}\int \ln x\, dx^4 = \frac{1}{2}x^4\ln x - \frac{1}{2}\int x^4 d\ln x$

$\int x^4 d\ln x = \cancel{x^4\ln x - \int \ln x\, d} \int x^3\,dx = \frac{x^4}{4}+C$

故 $\int x^4 d\ln^2 x = \frac{1}{2}x^4\ln x - \frac{x^4}{8}+C$

原式 $= \frac{1}{4}x^4\ln^2 x - \frac{1}{8}x^4\ln x + \frac{1}{32}x^4 + C.$

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17. $\int \frac{x \arctan x}{(1+x^2)^{5/2}} dx = \int \frac{x}{(1+x^2)^{3/2}} \arctan x \, d\arctan x$

$= \frac{1}{2} \int \frac{\arctan x}{(1+x^2)^{5/2}} d(1+x^2) = -\frac{1}{3} \int \arctan x \, d(1+x^2)^{-\frac{3}{2}}$

$= -\frac{1}{3} \frac{\arctan x}{(1+x^2)^{3/2}} + \frac{1}{3} \int \frac{1}{(1+x^2)^{3/2}} d\arctan x$

$= \int \frac{dx}{(1+x^2)^{5/2}}$ | 令 $x = \tan\theta$. $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $\sqrt{1+x^2} = \frac{1}{\cos\theta} = \sec\theta$, $dx = \sec^2\theta \, d\theta$

$\int \frac{dx}{(1+x^2)^{5/2}} = \int \frac{\sec\theta \, d\theta}{\sec^5\theta} = \int \cos^4\theta \, d\theta$

$= \int (\cos^2\theta)^2 d\theta = \int (\frac{1+\cos 2\theta}{2})^2 d\theta = \int \frac{1}{4} + \frac{1}{2}\cos 2\theta + \frac{1}{4}\cos^2 2\theta \, d\theta$ ($\frac{1+\cos 4\theta}{2}$)

$= \int \frac{3}{8} + \frac{1}{2}\cos 2\theta + \frac{1}{8}\cos 4\theta \, d\theta = \frac{3}{8}\theta + \frac{1}{4}\sin 2\theta + \frac{1}{32}\sin 4\theta + C$

$= \frac{3}{8}\arctan x + \frac{1}{2}\frac{x}{\sqrt{1+x^2}} \cdot \frac{1}{\sqrt{1+x^2}} -$

$\sin\theta = \frac{x}{\sqrt{1+x^2}}$

注意到 $\int \frac{dx}{(1+x^2)^{3/2}} = \frac{x}{(1+x^2)^{3/2}} - \int x \, d(1+x^2)^{-\frac{3}{2}}$

$= \frac{x}{(1+x^2)^{3/2}} + \frac{3}{2}\int \frac{2x^2}{(1+x^2)^{5/2}} dx$

$= \frac{x}{(1+x^2)^{3/2}} + 3\int \frac{1}{(1+x^2)^{3/2}} - \frac{1}{(1+x^2)^{5/2}} dx$

故 $\int \frac{dx}{(1+x^2)^{5/2}} = \frac{x}{3(1+x^2)^{3/2}} + \frac{2}{3}\int \frac{dx}{(1+x^2)^{3/2}}$

而 $\int \frac{dx}{(1+x^2)^{3/2}} \xlongequal{x=\tan\theta} \int \frac{\sec^2\theta \, d\theta}{\sec^3\theta} = \int \cos\theta \, d\theta = \sin\theta + C = \frac{x}{\sqrt{1+x^2}} + C$

故 $\int \frac{dx}{(1+x^2)^{5/2}} = \frac{x}{3(1+x^2)^{3/2}} + \frac{2x}{3\sqrt{1+x^2}} + C = \frac{2x^3+3x}{3(1+x^2)^{3/2}} + C$

$\Rightarrow$ 原式 $= \frac{1}{9(1+x^2)^{3/2}}(2x^3+3x-3\arctan x) + C$.

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18. $\int x \ln(x+\sqrt{1+x^2})\,dx = \int \ln(x+\sqrt{1+x^2})\,d\frac{x^2}{2}$

$= \frac{x^2}{2}\ln(x+\sqrt{1+x^2}) - \frac{1}{2}\underbrace{\int x^2\, d\ln(x+\sqrt{1+x^2})}_{I.}$

$I. = \int x^2 \cdot \frac{1}{\sqrt{1+x^2}}\,dx = \int x\, d\sqrt{1+x^2} = x\sqrt{1+x^2} - \int \sqrt{1+x^2}\,dx.$

$\int \sqrt{1+x^2}\,dx \overset{x=\tan\theta}{=\!=\!=} \int \sec\theta \cdot \sec^2\theta\, d\theta = \int \frac{d\theta}{\cos^3\theta}$

$\int \sqrt{1+x^2}\,dx \overset{\substack{x=\frac{e^t-e^{-t}}{2} \\ \sqrt{1+x^2}=\frac{e^t+e^{-t}}{2}}}{=\!=\!=} \int \frac{e^t+e^{-t}}{2}\, d\frac{e^t-e^{-t}}{2} = \frac{1}{4}\int (e^t+e^{-t})^2\,dt$

$= \frac{1}{4}\int (e^{2t}+e^{-2t}+2)\,dt = \frac{1}{8}(e^{2t}-e^{-2t}) + \frac{t}{2} + C$

$= \frac{1}{2}\cdot\frac{e^t+e^{-t}}{2}\cdot\frac{e^t-e^t}{2} + \frac{t}{2} + C.$

$e^{2t} - 2xe^t - 1 = 0 \Rightarrow e^t = \frac{2x \pm \sqrt{x^2+1}}{2} = x+\sqrt{x^2+1}$ (取正根)

$\Rightarrow t = \ln(x+\sqrt{x^2+1})$.

那么, $\int \sqrt{1+x^2}\,dx = \frac{1}{2}x\sqrt{1+x^2} + \frac{1}{2}\ln(x+\sqrt{x^2+1}) + C$

$\Rightarrow$ 原式 $= \frac{x^2}{2}\ln(x+\sqrt{1+x^2}) - \frac{1}{2}x\sqrt{1+x^2} + \frac{1}{4}x\sqrt{1+x^2} + \frac{1}{4}\ln(x+\sqrt{1+x^2}) + C$

$= \frac{x^2}{2}\ln(x+\sqrt{1+x^2}) + \frac{1}{4}\ln(x+\sqrt{1+x^2}) - \frac{1}{4}x\sqrt{1+x^2} + C$

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